

Notes: Bordalo, Gennaioli, and Shleifer (JF, 2018)
Diagnostic Expectations and Credit Cycles

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1 Big picture

- **Question.** How can credit booms, compressed credit spreads, neglected downside risk, and later reversals arise even in a model without standard financial frictions? The paper asks whether a psychologically founded expectations rule can generate credit cycles.
- **Core mechanism.** The model is built around *diagnostic expectations*: agents overweight future outcomes whose probability has risen the most in light of recent news. After good news, good future states become diagnostic, so investors become too optimistic and neglect downside risk.
- **Psychological foundation.** Diagnostic expectations formalize Kahneman and Tversky’s representativeness heuristic. An outcome looks likely when it is especially representative of the current situation relative to a reference situation.
- **Forward-looking feature.** Unlike mechanical extrapolation, diagnostic expectations depend on the true stochastic process. If fundamentals are not persistent, there is little reason to extrapolate; if fundamentals are persistent, recent news is informative and therefore becomes diagnostic.
- **Credit-market implication.** Good news reduces perceived default risk, compresses credit spreads, expands debt issuance, increases investment, and raises the share of risky debt. But the initial optimism is excessive, so spreads later reverse upward.
- **Main empirical motivation.** Blue Chip forecasts of Baa-Treasury credit spreads contain predictable errors and predictable revisions. When spreads are low, forecasts are too optimistic and later revised upward; when spreads are high, forecasts are too pessimistic and later revised downward.
- **Main theoretical result.** Under diagnostic expectations, equilibrium credit spreads follow an ARMA-type process rather than a simple AR(1). The moving-average component captures predictable unwinding of earlier overreaction.
- **Economic takeaway.** Credit booms contain the seeds of later busts because beliefs overreact to genuinely informative good news. The model reconciles extrapolation, neglect of risk, excess volatility of credit spreads, predictable returns, and macroeconomic reversals in one framework.

2 Incremental contribution of the paper

- **From narrative credit cycles to a microfounded expectations rule.** The paper gives a formal expectations mechanism behind Minsky-style boom-bust narratives. Optimism is not simply assumed; it follows from representativeness-based belief formation.
- **A bridge between psychology and macro-finance.** The same representativeness logic used to explain laboratory judgment errors is applied to forecasts of aggregate fundamentals and credit spreads.
- **Unifies extrapolation and neglected risk.** The model shows that the same belief distortion can produce both extrapolative optimism and neglect of downside states. After good news, the right tail is overweighted and the left tail is underweighted.

- **Forward-looking behavioral expectations.** The paper improves on mechanical extrapolation because the diagnostic distortion depends on the underlying data-generating process. This makes the rule responsive to persistence, volatility, and policy changes.
- **Credit-cycle predictions beyond rational expectations.** Rational expectations can produce lower spreads and higher investment after good news, but it does not naturally deliver predictable forecast errors, predictable forecast revisions, excess spread volatility, and abnormal bond returns. Diagnostic expectations do.
- **A simple macro model without relying on financial frictions.** The baseline model deliberately avoids agency frictions, collateral constraints, or leverage amplification. The paper shows that belief dynamics alone can generate many credit-cycle facts.
- **An extension to safe debt and fragility.** The preferred-habitat extension connects belief distortions to the special role of safe debt. Excess optimism can lead risky debt to be temporarily classified as safe, making later reclassification especially destabilizing.
- **A calibration contribution.** Using the Blue Chip forecast regressions, the paper gives a back-of-the-envelope calibration of the diagnosticity parameter, finding a value around one. This disciplines the behavioral parameter with observed expectations data.

3 Data and measurement

3.1 Blue Chip forecast data

- The motivating evidence uses *Blue Chip Financial Forecasts*, a monthly survey of roughly 40 forecasters from major financial institutions.
- The authors use the consensus forecast, defined as the mean forecast across panelists.
- The sample uses the March, June, September, and December surveys, so the observations are quarterly.
- The forecast variable is the forecast of the Baa corporate bond yield minus the forecast of the 10-year Treasury yield.
- The sample period is 1999Q1 through 2014Q4.

Interpretation: The expectations data are not used as a full structural estimation sample. They are used to motivate the model and to discipline the size of forecast errors and revisions.

3.2 Credit-spread variables

- The main credit-spread object is the Baa corporate bond spread over the 10-year Treasury yield.
- The paper uses the past four-quarter average of the actual spread as the current credit-market condition.
- The realized future spread is the actual spread averaged over quarters $t + 1$ to $t + 4$.
- The forecasted future spread is the quarter- t forecast for the spread averaged over quarters $t + 1$ to $t + 4$.

- The forecast error is defined as actual minus forecast.

Interpretation: A negative coefficient of forecast errors on current spreads means that when current spreads are low, future realized spreads exceed forecasts. This is the empirical signature of excessive optimism in boom periods.

3.3 Forecast revisions

- The paper also studies revisions in expectations.
- The revision variable compares the forecast made in quarter $t + 3$ for the spread in quarter $t + 4$ with the forecast made in quarter t for that same quarter- $t + 4$ spread.
- Under rational expectations, forecast errors and forecast revisions should not be predictable from information already available at time t .
- The evidence shows that revisions are predictable from the current spread.

Interpretation: The revision test is important because it captures the dynamic correction of beliefs. If forecasters are too optimistic when spreads are low, they should later revise spreads upward. This is exactly the pattern the paper documents.

3.4 Model primitives treated as measured objects

The paper is primarily a theory paper, but the empirical objects map closely into the model.

- The fundamental state is ω_t , interpreted as aggregate economic conditions.
- Credit spreads are inverse measures of optimism: high perceived creditworthiness lowers spreads.
- A positive shock ε_t raises expected future fundamentals and compresses spreads.
- The diagnosticity parameter θ measures the strength of belief distortion. When $\theta = 0$, expectations are rational. When $\theta > 0$, agents overweight diagnostic states.

Interpretation: The empirical spread regressions are not a full test of the whole model. They establish the belief-reversal facts that the model is designed to explain.

4 Models and empirical specifications

4.1 Representativeness and diagnostic beliefs

The paper starts from the idea that an outcome is representative of a group when it is relatively more frequent in that group than in a comparison group. If G is the current group and $-G$ is the reference group, representativeness is

$$\frac{h(T = t \mid G)}{h(T = t \mid -G)}.$$

- In the paper's static example, red hair is representative of the Irish because red hair is much more frequent among the Irish than in the world at large.

- In the macro-finance application, the group is the current state of the economy and the comparison group is the no-news state implied by yesterday's information.
- An event becomes diagnostic when its probability increases sharply after new information arrives.

Interpretation: Diagnostic expectations are not random mistakes. They contain a kernel of truth: agents react to genuinely informative news, but they overreact to the states whose probability rises the most.

4.2 Dynamic expectation formation

Let fundamentals follow an AR(1) process:

$$\omega_t = b\omega_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2), \quad b \in [0, 1].$$

At time t , the rational conditional distribution of ω_{t+1} is $h(\omega_{t+1} | \omega_t)$. The diagnostic comparison distribution is the no-news distribution $h(\omega_{t+1} | b\omega_{t-1})$. The distorted diagnostic density is

$$h_t^\theta(\omega_{t+1}) = h(\omega_{t+1} | \omega_t) \left[\frac{h(\omega_{t+1} | \omega_t)}{h(\omega_{t+1} | b\omega_{t-1})} \right]^\theta \frac{1}{Z},$$

where Z normalizes the density.

- $\theta = 0$ gives rational expectations.
- $\theta > 0$ inflates the likelihood of states that became more likely after the current news.
- Good news shifts diagnostic beliefs toward high future states.
- Bad news shifts diagnostic beliefs toward low future states.

For the normal AR(1) case, Proposition 1 gives a simple representation:

$$E_t^\theta(\omega_{t+1}) = E_t(\omega_{t+1}) + \theta [E_t(\omega_{t+1}) - E_{t-1}(\omega_{t+1})].$$

Interpretation: The formula says that diagnostic expectations equal rational expectations plus an overreaction term. Agents do not mechanically extrapolate every movement; they overweight news only to the extent that the news changes the true conditional distribution.

4.3 Extrapolation and neglect of risk

For the AR(1) process, the diagnostic forecast can be written as

$$E_t^\theta(\omega_{t+1}) - \omega_t = [E_t(\omega_{t+1}) - \omega_t] + b\theta [\omega_t - E_{t-1}(\omega_t)].$$

- The last term shows extrapolation of the current shock.
- The amount of extrapolation is proportional to persistence b .
- If $b = 0$, the current shock is not informative about the future, so diagnostic expectations do not mechanically extrapolate it.

- Because the whole distribution is shifted after good news, downside states receive too little probability. This is neglect of downside risk.
- Neglect of tail risk is especially severe when volatility σ^2 is low, because even moderate news can strongly change the relative likelihood of tail states.

Interpretation: This is the paper's central behavioral synthesis. Extrapolation and neglected risk are not separate assumptions; they are two implications of the same representativeness-based updating rule.

4.4 Production and default model

A continuum of firms uses capital to produce output. Firms differ by risk type ρ . A firm produces only if the aggregate state is high enough:

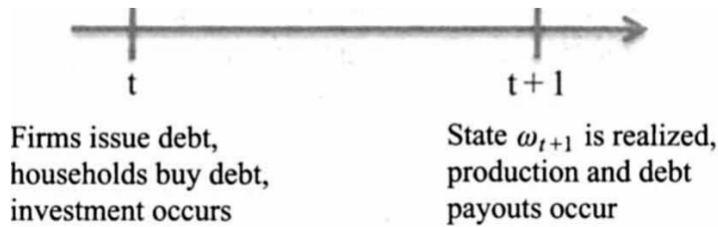
$$y(k \mid \omega_t, \rho) = \begin{cases} k^\alpha, & \text{if } \omega_t \geq \rho, \\ 0, & \text{if } \omega_t < \rho. \end{cases}$$

- Higher ρ means a riskier firm because the firm requires a better aggregate state to repay.
- Capital for production at $t + 1$ is installed at t before ω_{t+1} is realized.
- Firms borrow by issuing debt that promises an interest rate $r_{t+1}(\rho)$.
- Debt is repaid only if the firm produces, that is, if $\omega_{t+1} \geq \rho$.

The perceived repayment probability is

$$\mu(\rho, E_t^\theta(\omega_{t+1})) = \int_\rho^\infty h_t^\theta(x) dx.$$

Interpretation: Diagnostic expectations enter the real economy through perceived default probabilities. When investors are optimistic, they perceive risky firms as safer, and the cost of risky credit falls.



Model timeline extracted from the paper: firms issue debt and invest at t ; the state is realized and production and debt payouts occur at $t + 1$.

4.5 Capital-market equilibrium and spreads

Firms demand capital by equating marginal product and promised repayment cost:

$$k_{t+1}(\rho) = \left[\frac{\alpha}{r_{t+1}(\rho)} \right]^{\frac{1}{1-\alpha}}.$$

Households are risk neutral and require expected return β^{-1} :

$$r_{t+1}(\rho) \cdot \mu\left(\rho, E_t^\theta(\omega_{t+1})\right) = \beta^{-1}.$$

Therefore, the risky debt spread is

$$S\left(\rho, E_t^\theta(\omega_{t+1})\right) = \left(\frac{1}{\mu(\rho, E_t^\theta(\omega_{t+1}))} - 1\right) \beta^{-1}.$$

Combining the pricing and investment conditions gives

$$k_{t+1}(\rho) = \left[\frac{\alpha\beta}{1 + \beta S(\rho, E_t^\theta(\omega_{t+1}))} \right]^{\frac{1}{1-\alpha}}.$$

- Higher perceived creditworthiness raises μ .
- Higher μ lowers spreads.
- Lower spreads raise investment.
- Riskier firms' spreads and investment respond more strongly to aggregate optimism.

Interpretation: The baseline credit boom is simple: good news creates optimism, optimism lowers perceived default risk, lower perceived default risk compresses spreads, and compressed spreads expand borrowing and investment.

4.6 Equilibrium spread dynamics

The paper linearizes the average spread around the long-run mean:

$$S_t \approx \sigma_0 - \sigma_1 E_t^\theta(\omega_{t+1}), \quad \sigma_0, \sigma_1 > 0.$$

Substituting diagnostic expectations into the spread equation yields the key ARMA form:

$$S_t = (1 - b)\sigma_0 + bS_{t-1} - (1 + \theta)b\sigma_1\varepsilon_t + \theta b^2\sigma_1\varepsilon_{t-1}.$$

- Under rational expectations, $\theta = 0$, spreads follow an AR(1) driven by fundamentals.
- Under diagnostic expectations, $\theta > 0$, spreads contain a moving-average reversal term.
- Good news at $t - 1$ compresses spreads too much at $t - 1$.
- When that overreaction fades, spreads rise at t even without new bad fundamentals.

Interpretation: The ARMA term is the mathematical form of “boom-bust” credit-market sentiment. Crises can occur when earlier optimism unwinds, not only when fundamentals deteriorate.

4.7 Forecasts, forecast errors, and revisions

Diagnostic forecasters project current spreads forward according to

$$E_t^\theta(S_{t+T}) = \sigma_0(1 - b^T) + b^T S_t.$$

But the realized equilibrium spread contains the reversal term that forecasters fail to anticipate. Hence forecast errors and revisions are predictable:

$$E_t \left[S_{t+1} - E_t^\theta(S_{t+1}) \right] = \theta b^2 \sigma_1 \varepsilon_t,$$

$$E_t \left[E_{t+s}^\theta(S_{t+T}) - E_t^\theta(S_{t+T}) \right] = \theta b^{T+1} \sigma_1 \varepsilon_t.$$

- After good news, $\varepsilon_t > 0$, spreads are too low and future realized spreads exceed forecasts.
- Future forecasts are revised upward after good news.
- After bad news, the signs reverse: spreads are too high and future forecasts are revised downward.

Interpretation: These equations are the theoretical counterpart of Figures 1 and 2. Low current spreads predict positive future forecast errors and upward forecast revisions.

4.8 Returns, excess volatility, and real activity

Let S_t^r denote the spread that would prevail under rational expectations. Diagnostic expectations imply

$$S_t - S_t^r = -\theta b \sigma_1 \varepsilon_t.$$

- After good news, spreads are below rational compensation for default risk, so bond returns are predictably low.
- After bad news, spreads are above rational compensation, so future returns are predictably high.
- Conditional spread volatility is excessive:

$$Var(S_t \mid \omega_{t-1}) = (1 + \theta)^2 Var(S_t^r \mid \omega_{t-1}).$$

- Low spreads caused by recent optimism predict future spread widening, reduced investment, and lower future output.

Interpretation: The model links financial-market sentiment to real activity. Belief reversals tighten credit conditions, which reduces investment and output even if the original news was good.

4.9 Safe debt extension

The extension introduces a preferred habitat for safe debt. Investors require lower returns for claims whose perceived default probability is below a threshold δ^* .

- A threshold risk type ρ_t^θ solves

$$\mu(\rho_t^\theta, E_t^\theta(\omega_{t+1})) = 1 - \delta^*.$$

- Firms with $\rho < \rho_t^\theta$ issue debt perceived as safe.
- Firms with $\rho > \rho_t^\theta$ issue risky debt and must offer a higher expected return.
- Good news raises perceived safety and expands the set of assets classified as safe.
- When optimism unwinds, marginally safe assets can be reclassified as risky, causing a sharp price drop.

Interpretation: Preferred habitat turns a smooth belief mistake into financial fragility. The most fragile securities are those that were only safe because investors temporarily neglected risk.

5 Identification and empirical design

5.1 What the motivating evidence establishes

- The Blue Chip regressions establish predictability in credit-spread forecast errors and forecast revisions.
- They show that professional forecasters appear too optimistic in low-spread periods and too pessimistic in high-spread periods.
- These facts are difficult to reconcile with rational expectations because rational forecast errors and revisions should not be predictable using information available at the forecast date.

Interpretation: The empirical section is not a complete structural test of the model. Its purpose is to document belief dynamics that the model can explain and rational-expectations models struggle to explain.

5.2 What the model identifies conceptually

- The model identifies how an expectations distortion alone can generate credit cycles.
- Because the baseline has no financial frictions, any excess volatility, predictable reversals, and abnormal returns come from beliefs rather than balance-sheet constraints.
- The safe-debt extension then shows how financial frictions or preferred-habitat demand can amplify the belief-driven cycle.

Interpretation: The clean baseline is useful because it separates the source of shocks from the amplification mechanism. Beliefs create the shock; leverage and safe debt can make it more dangerous.

5.3 Calibration of diagnosticity

- The paper uses the Table I forecast-error regression to obtain a back-of-the-envelope calibration of θ .
- The persistence of forecasted spreads is estimated from the regression of forecasts on current spreads.

- The forecast-error coefficient is then mapped into the model’s expression for forecast-error predictability.
- The resulting diagnosticity parameter is approximately $\theta = 0.91$.

Interpretation: A value near one means that investors’ diagnostic overreaction is quantitatively large: forecast mistakes are of the same order as actual spread innovations.

5.4 Strengths of the design

- The paper uses actual expectations data rather than inferring all beliefs from prices.
- The expectation facts directly match the theoretical objects of forecast errors and revisions.
- The model generates a connected set of predictions: spreads, credit quantities, risky debt share, bond returns, investment, output, and safe-debt fragility.
- The expectations rule is portable because it comes from a general psychology mechanism rather than a credit-market-specific reduced form.

Interpretation: The paper’s strength is coherence: one belief mechanism explains many facts that otherwise require separate assumptions.

5.5 Limitations and interpretation

- The Blue Chip sample is short, covering roughly 64 quarterly observations.
- The forecast evidence is motivating rather than definitive; it cannot by itself rule out every rational explanation involving time-varying risk premia or omitted state variables.
- The model is stylized: firms differ by one risk parameter, investors are representative in the baseline, and financial frictions are introduced only in an extension.
- The baseline deliberately abstracts from institutional details of credit markets.

Interpretation: The right reading is that the paper offers a disciplined behavioral mechanism for credit cycles, not that it fully estimates the entire credit market.

6 Tables and figures

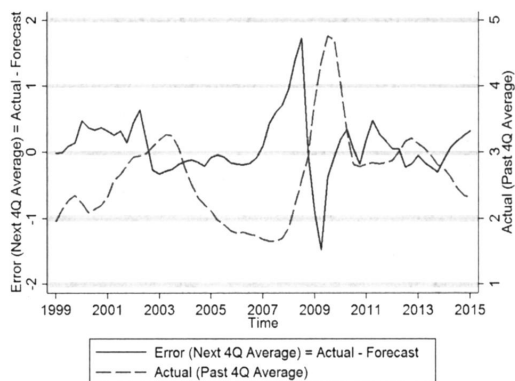
6.1 Figure 1 and Table I: Forecast errors of credit spreads

Read:

- Figure 1 plots the current credit spread against forecast errors for the next four-quarter average spread.
- The solid line is the forecast error, actual minus forecast. The dashed line is the past four-quarter average spread.
- Low-spread periods such as the mid-2000s are associated with positive future forecast errors: realized spreads later exceed forecasts.

- Table I formalizes the pattern in regressions. The current spread coefficient is 0.39 for future actual spreads, 0.65 for forecasts, and -0.26 for forecast errors.
- The forecast coefficient exceeds the actual coefficient, meaning forecasters perceive too much persistence in current credit conditions.

Economic message: Table I is the first empirical anchor for diagnostic expectations. When credit markets look safe, forecasters behave as if safety will persist too strongly. The later correction shows up as predictable forecast errors.



(a) Figure 1: Predictable errors in forecasts of credit spreads.

Table I
Actual, Forecast, and Error of Future Credit Spreads

Quarterly time series regressions: the dependent variable is the actual credit spread averaged over quarters $t + 1$ to $t + 4$ in column (1), quarter t forecasts of credit spreads averaged over quarters $t + 1$ to $t + 4$ in column (2), and the forecast error (actual–forecast) of credit spreads in column (3); the independent variable is the actual credit spread averaged over quarters $t - 4$ to $t - 1$, where $t - 1$ is the latest quarterly credit spread prior to the forecast. Standard errors in parentheses are Newey-West, with the automatic bandwidth selection of Newey and West (1994).

	Actual (1)	Forecast (2)	Error (3)
Current spread	0.39 (1.67)	0.65 (4.62)	-0.26 (-2.20)
Constant	1.63 (2.56)	0.86 (2.25)	0.77 (2.40)
Observations	64	64	64
R^2	0.16	0.47	0.16

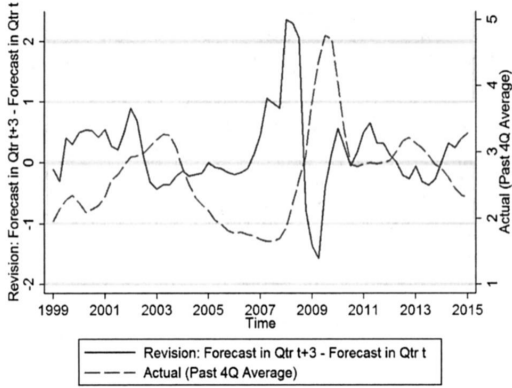
(b) Table I: Actual, forecast, and error of future credit spreads.

6.2 Figure 2 and Table II: Forecast revisions of credit spreads

Read:

- Figure 2 plots future forecast revisions against the current credit spread.
- The revision compares the forecast made in quarter $t + 3$ with the forecast made in quarter t for the same quarter- $t + 4$ spread.
- When the current spread is low, forecasts tend to be revised upward later.
- When the current spread is high, forecasts tend to be revised downward later.
- Table II shows a coefficient of -0.36 of revisions on the current spread, with a negative and statistically meaningful relationship.

Economic message: The revision evidence is especially important because it documents the unwinding of expectations. Investors do not merely make noisy errors; they systematically correct earlier optimism or pessimism.



(a) Figure 2: Predictable revisions in forecasts of credit spreads.

Table II
Forecast Revisions of Credit Spreads

Quarterly time series regression: the dependent variable is the forecast revision (quarter $t+3$ forecast of credit spread in quarter $t+4$ minus quarter t forecast of credit spread in quarter $t+4$); the independent variable is actual credit spread averaged over quarters $t-4$ to $t-1$, where $t-1$ is the latest quarterly credit spread prior to the forecast. Standard errors in parentheses are Newey-West, with the automatic bandwidth selection of Newey and West (1994).

	Revision (1)
Current spread	-0.36 (-2.13)
Constant	1.13 (2.44)
Observations	64
R^2	0.15

(b) Table II: Forecast revisions of credit spreads.

6.3 Figure 3: Neglect of risk and extrapolation

Read:

- Figure 3 shows how diagnostic expectations transform the distribution of future fundamentals after good news.
- The lagged rational distribution is on the left, the current rational distribution is in the middle, and the diagnostic distribution shifts further to the right.
- The shaded left-tail region is the downside risk that is underweighted after good news.
- The same shift creates both extrapolation of good news and neglect of left-tail risk.
- This visual is the paper's key conceptual diagram: diagnostic optimism is a distributional distortion, not just a higher point forecast.

Economic message: The figure explains why tranquil booms can be dangerous. When good news arrives in a low-volatility environment, investors may only mildly overstate the mean but severely understate tail risk.

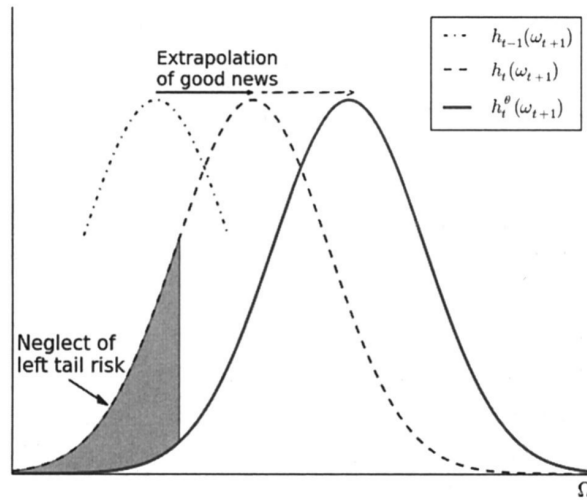


Figure 3: Neglect of risk and extrapolation.

7 Short summary

- The paper develops diagnostic expectations, a forward-looking belief rule based on representativeness.
- Agents overweight future states whose probabilities rise most after current news.
- After good news, agents become too optimistic and neglect downside risk.
- The model is not mechanical extrapolation: the degree of extrapolation depends on the persistence and volatility of fundamentals.
- Blue Chip forecasts of Baa-Treasury spreads show predictable errors and predictable revisions.
- In the model, credit spreads are excessively volatile and follow an ARMA process because overreaction later unwinds.
- Low spreads caused by optimism predict later spread widening, lower credit, lower investment, and weaker output.
- Investors earn predictably low returns after good news because spreads undercompensate for default risk.
- The safe-debt extension shows how neglect of risk can create fragility when risky debt is temporarily treated as safe.
- The paper’s main contribution is to provide a psychologically grounded, forward-looking theory of credit cycles.

8 Conclusion

8.1 Core conclusion

Bordalo, Gennaioli, and Shleifer show that credit cycles can arise from a systematic and psychologically grounded distortion in expectations. Good news does not merely improve fundamentals; it also changes which future states come to mind. Investors overweight the states that became more likely and underweight states that became less representative. In credit markets, this means excessive optimism, compressed spreads, expanded credit, and neglected downside risk. When the optimism unwinds, spreads rise and credit conditions tighten.

8.2 Economic intuition

The intuition is easiest to see in a boom. A sequence of good news makes strong future states seem representative of the economy. Investors therefore think default is less likely than it really is, especially for risky borrowers. Credit spreads fall too much, risky debt issuance rises, and investment expands. But the initial belief distortion is temporary. Even if fundamentals do not collapse, the overreaction fades. The resulting spread widening reduces investment and can generate a downturn.

8.3 Incremental contribution revisited

The paper advances the credit-cycle literature in three main ways. First, it gives a microfounded behavioral expectation rule rather than assuming generic optimism. Second, it unifies extrapolation and neglected risk in a single distributional mechanism. Third, it embeds that mechanism in a credit-market model that produces forecast errors, forecast revisions, excess volatility, abnormal returns, and macroeconomic reversals.

8.4 Implications for finance and macroeconomics

The paper implies that credit-market sentiment is not just noise. It can move prices, debt issuance, investment, and future output. Low spreads are not always a sign of low risk; they may reflect excessive optimism. The model also suggests that stable periods can be fragile because low volatility strengthens the diagnostic neglect of tail risk. For policy, the paper supports paying attention to credit-market frothiness, compressed spreads, and growth in risky debt issuance.

8.5 Limitations

The paper is intentionally stylized. It does not fully estimate a structural model of credit markets, and the expectations evidence is based on a relatively short survey sample. The baseline model abstracts from many institutional features, including intermediaries, collateral constraints, heterogeneous investors, and regulatory details. The safe-debt extension adds amplification but remains simple. These limitations are also strengths in one sense: the paper isolates the belief channel before adding frictions.

8.6 Takeaway

The enduring takeaway is that credit booms can be driven by a subtle form of overreaction to real information. Good news is informative, but investors treat the most diagnostic future outcomes as too likely. That creates extrapolation, neglect of risk, compressed spreads, and later reversals. The paper turns the Minsky-style idea that optimism breeds fragility into a formal, portable, and testable expectations framework.